

BENCHMARK: MARGINAL

Results

|                                                   |                                  |
|---------------------------------------------------|----------------------------------|
| Mid-Transit Time (Tmid)                           | 2458608.6827 +/- 0.00094 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.198 +/- 0.011                  |
| Transit depth (Rp/Rs)^2                           | 3.9 +/- 0.45 [%]                 |
| Orbital Inclination (inc)                         | 79.71 +/- 0.4                    |
| Airmass coefficient 1 (a1)                        | 0.9621 +/- 0.0095                |
| Airmass coefficient 2 (a2)                        | 0.0243 +/- 0.0064                |
| Scatter in the residuals of the lightcurve fit is | 0.98 %                           |
| Best Comparison Star                              | #3 - [407, 315]                  |
| Optimal Aperture                                  | 4.04                             |
| Optimal Annulus                                   | 9.47                             |
| Transit Duration (day)                            | 0.0448 +/- 0.0021                |

Accuracy vs published ephemeris (ExoClock/NEA)

|                                      |                                                        |
|--------------------------------------|--------------------------------------------------------|
| Rp/R $\blacksquare$ fit vs published | 0.198 $\pm$ 0.011 vs 0.1594 (deviation 2.56 $\sigma$ ) |
| Duration fit vs published            | 0.0448 d vs 0.0512 d (deviation 2.24 $\sigma$ )        |
| Mid-time O-C                         | 1.7 min                                                |
| Depth SNR                            | 13.2                                                   |
| Residual scatter                     | 0.98 %                                                 |

Light curve

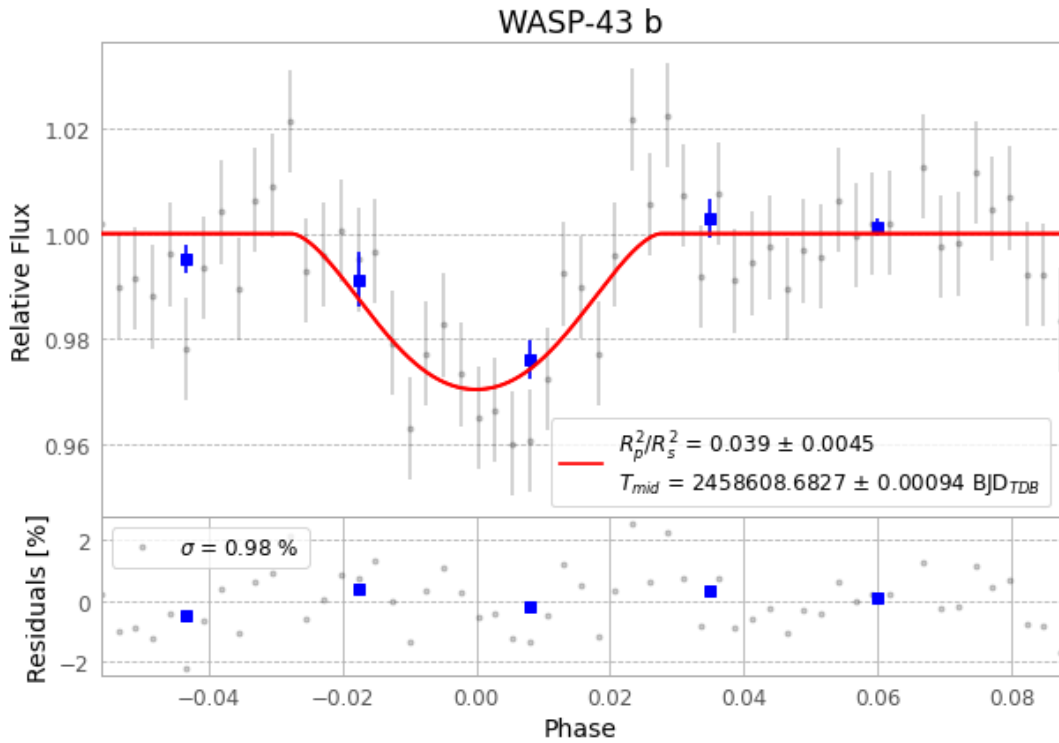


Figure 1 — FinalLightCurve\_WASP-43 b\_2019-05-04.png (SHA-256 730b15e404e5e503...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [385, 187], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 b61f5dfb382e7942...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ba3ef8b

## Data provenance

57 FITS frames (37 MB), WASP-43190505031212.FITS → WASP-43190505060012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-190505.log (c4428ceafec0...), FinalLightCurve\_WASP-43 b\_2019-05-04.png (730b15e404e5...), FinalLightCurve\_WASP-43 b\_2019-05-04.csv (60e4533ee11f...)

## Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 07:56Z.